

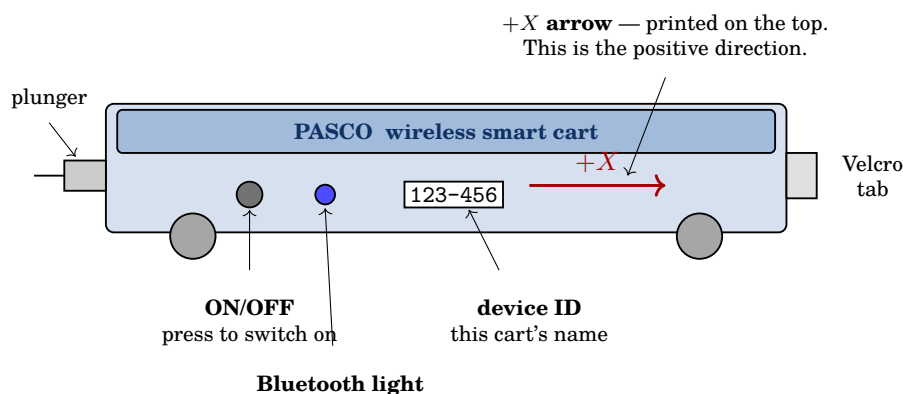
Connecting SPARKvue

Getting a PASCO Smart Cart talking to a laptop — and what to do when it will not.

Key concept

The one-line version. Switch the cart on, check its Bluetooth light is **blinking red**, open SPARKvue, choose **Sensor Data**, and pick the cart whose **device ID** matches the number printed on the cart in your hands.

Know the cart



Schematic — your cart is red or blue and the parts sit in slightly different places, but every one of these is on it somewhere.

The Bluetooth light is the whole diagnostic story. Learn it and you will almost never need to ask for help. (Charging is not your problem — the carts are charged for you.)

Light	What it does	What it means
Bluetooth	● blinking red	Switched on, ready to pair . This is what you want to see before you open SPARKvue.
Bluetooth	● blinking green	Connected to the software. You are ready to record.
Bluetooth	nothing	Either the cart is off, or it is plugged in by USB — the Bluetooth light stays dark on a USB connection.

Connecting, step by step

1. **Press the ON button on the cart.** Hold it briefly if nothing happens. Check the **Bluetooth light is blinking red**. If it is not, the cart is not on.
2. **Read the device ID off your cart.** It is a **six-digit number**, like 123-456. Write it on your hand if you have to — in a room of eight carts, this is the only thing that tells yours apart.
3. **Open SPARKvue on the laptop.**
4. **From the Welcome screen, choose *Sensor Data*.**
5. **Click *Bluetooth Devices*, top right.** SPARKvue lists what it can see, **nearest first**.
6. **Pick the cart whose ID matches yours.** Not the top of the list — the one that matches.
7. **Tick the measurements you want** in the *Select Measurements for Templates* list. For every lesson this term that is **Position** and **Velocity**. Leave Force and the three accelerations unticked; they only clutter the graph.
8. **In *Templates*, choose *Graph*.** SPARKvue builds the graph for you, with what you ticked up the side and time along the bottom.
9. **Press *Start*** when you are ready to record, and *Stop* when the run is over.

If your SPARKvue looks different. Newer versions connect from the **Bluetooth button on the top toolbar** instead. Click it to open **Wireless Devices**, then **New Bluetooth Device**, choose the cart with your device ID, then **Pair**, then **Done**. Same idea, different door — you are still matching the printed ID.

What the cart can actually measure. Position and velocity come from the encoder wheels, so they only work when it is rolling on its wheels. It also measures force (up to ± 100 N) and acceleration in three directions, and it weighs about 250 g. For the kinematics unit you need position and velocity and nothing else.

*Technical details taken from the PASCO Wireless Smart Cart Product Guide (012-14727D) and the SPARKvue help pages. Next: **Using SPARKvue** — what to do once the cart is connected.*